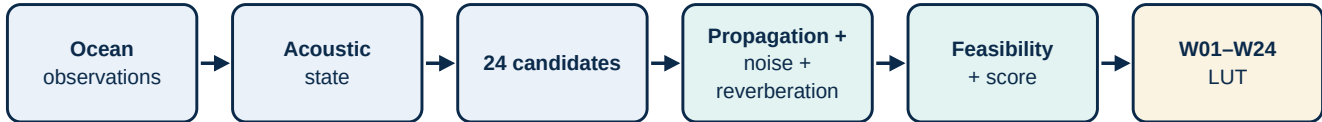


OJAS

Adaptive Ocean–Acoustic Waveform Selection

Technical and Scientific Report – SIH 2026



Core idea. OJAS does not use one fixed sonar waveform for every ocean condition. It converts the current environmental state into an acoustic state, evaluates a fixed library of 24 candidate waveforms, rejects candidates that fail hard limits, and selects the best feasible one. The expensive search is done offline and the final decision is stored as a compact deployment lookup representation.

Item	Final implementation
Source observations	$\approx 1,839$ ocean rows
Range scenarios	12 target ranges, 10–250 m
Waveform candidates	24: LFM, GSFM, Barker phase-coded
Offline evaluations	$1,839 \times 12 \times 24 = 529,632$
Absorption model	François–Garrison
Turbidity model	Volume-reverberation proxy
Runtime path	Arduino input $\rightarrow$ STM32 $\rightarrow$ laptop
Validation	MATLAB 10-second random-row replay

Table 1: Summary of the final OJAS implementation.

Prepared from the supplied OJAS generator, methodology/provenance files, MATLAB validation materials, deployment artifacts, and final judges script.

1. Problem, Goal and Design Idea

The design problem behind OJAS is simple: underwater acoustic conditions are not constant. Temperature, salinity, depth, turbidity, range, ambient noise and operating frequency all affect the useful performance of a transmitted pulse. A waveform that is good in one condition may not be the best choice in another.

OJAS therefore treats waveform choice as a decision problem rather than a fixed transmitter setting. The ocean data does not directly say “use waveform X”. Instead, it defines the environmental state. That state is passed through an acoustic model and a candidate library, and the system selects the best feasible waveform.

1.1 System objective

For an environment/scenario vector

$$\mathbf{e} = [D, T, S, \tau_{\text{turb}}, c, R, TS, W] \quad (1)$$

where D is depth, T temperature, S salinity, τ_{turb} turbidity, c sound speed, R target range, TS target strength and W the 1-kHz wind-noise reference, OJAS evaluates candidates $k = 1, \dots, 24$. Each candidate produces a performance vector containing absorption, transmission loss, received level, noise, reverberation, processing gain, detection metric, range resolution, sidelobes, correlation and estimated energy. The decision is

$$k^* = \arg \max_{k \in F(\mathbf{e})} J_k(\mathbf{e}) \quad (2)$$

where F contains only candidates that pass all hard constraints.

1.2 Why the LUT approach?

The full calculation is too large to repeat on a small embedded processor whenever the environment changes. OJAS therefore follows an offline-to-runtime pattern: ocean data → physics/modeling → optimization → compact LUT. At runtime the controller only needs the environmental input, the deployment lookup and the waveform definition.

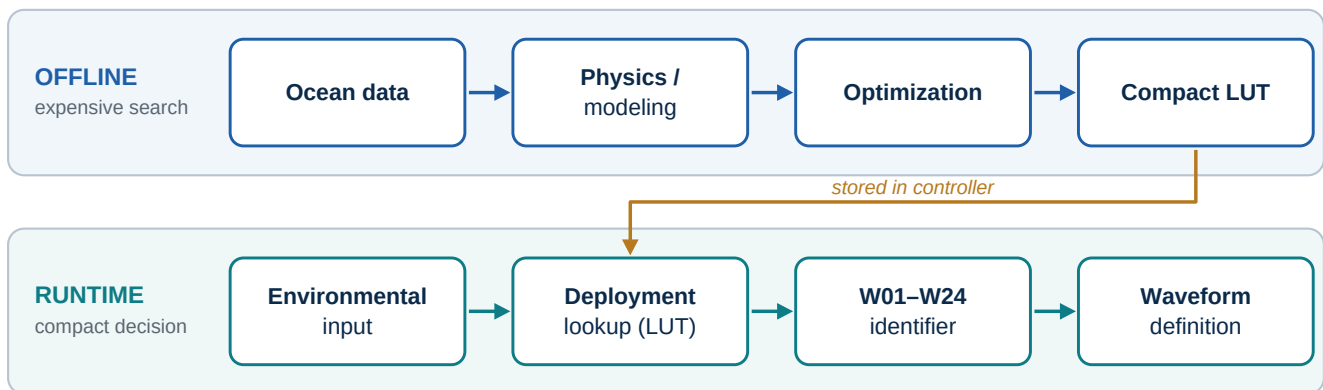


Figure 1: Offline-to-runtime pattern. The expensive search is completed offline; only the compact LUT and waveform definitions are used at runtime.

1.3 End-to-end architecture

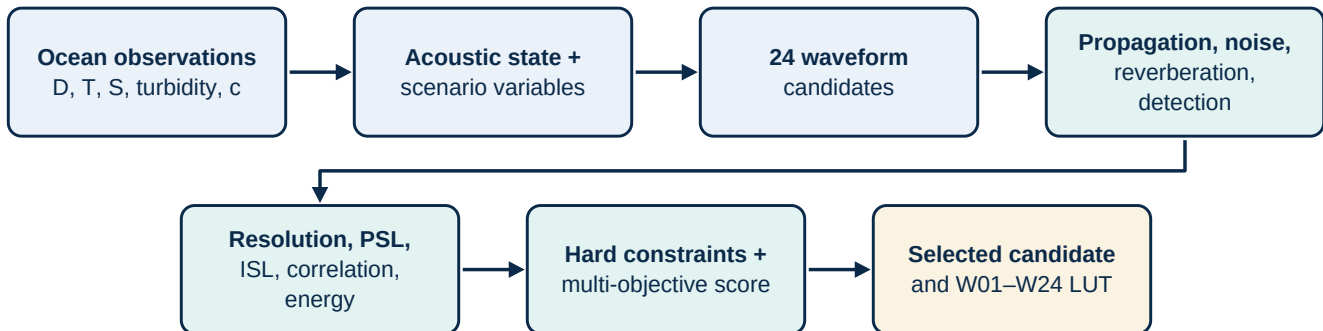


Figure 2: OJAS decision flow.

Important reporting rule: source environmental variables are recorded/supplied values. Acoustic performance values produced by the model are model-derived or engineering estimates.

2. Input Data, Provenance and Scenario Generation

2.1 Supplied environmental variables

The source dataset contains approximately 1,839 observations. The project methodology identifies the following as supplied/recorded inputs:

Field	Role
depth_m	Observation depth.
temperature_C	Temperature for sound-speed/absorption modeling.
salinity_PSU	Salinity for sound-speed/absorption modeling.
turbidity_NTU	Input to the volume-reverberation model.
sound_speed_mps_mackenzie	Supplied Mackenzie sound-speed field used as c .
sound_speed_data_mps	Additional supplied sound-speed field.
sound_speed_delta_mps	Difference between supplied sound-speed fields.
profile_depth_max_m	Profile context available in the dataset.

Table 2: Supplied environmental fields.

2.2 Derived quantities and assumptions

Not measured directly: approximate pressure, candidate absorption, transmission loss, received level, in-band noise, reverberation, processing gain, SINR, detection margin, range resolution, correlation, PSL/ISL and normalized energy. Key assumptions in the final generator are:

Parameter	Value	Purpose
Target strength	−25 to 0 dB	Scenario variation
Wind noise @ 1 kHz	45–65 dB	Scenario variation
Required resolution	0.04–0.08 m	Scenario requirement
Source level	180 dB re 1 μ Pa @ 1 m	Modeling assumption
Receiver noise figure	10 dB	Receiver/electronics assumption
pH	8.05	Needed by absorption model; not in source CSV
Minimum detection	10 dB	Processed SINR threshold

Table 3: Key modeling assumptions in the final generator.

2.3 Scenario expansion

The final fixed generator uses 12 target ranges generated by geometric spacing from 10 m to 250 m. Each ocean row is repeated for all 12 ranges. Target strength, wind-noise reference and maximum allowable range resolution are sampled with a fixed random seed of 42. Thus,

$$1,839 \times 12 = 22,068 \quad (3)$$

scenario rows are formed before candidate evaluation. All 24 candidates are then evaluated:

$$22,068 \times 24 = 529,632. \quad (4)$$

This large offline search is later compressed into the deployment decision structure.

3. Waveform Candidate Library

OJAS uses a fixed 24-candidate design space. The 24 candidates are not meant to represent every waveform possible; they are a manageable engineering set that covers three families, multiple frequencies, bandwidths and pulse durations.

3.1 Candidate families

- **LFM:** linear frequency modulation.
- **GSFM:** spectrally shaped nonlinear FM using a Hamming-weighted group-delay construction.
- **Phase-coded:** Barker-7 and Barker-13 sequences.

The center-frequency set is

$$f_c \in \{60, 100, 150, 200, 300, 400, 500, 590\} \text{ kHz},$$

and the bandwidth set is between 10 and 20 kHz.

3.2 Compact candidate map

ID range	Family	f_c set (kHz)	B trend (kHz)	T_p trend (ms)	Code
C01–C08	LFM	60, 100, 150, 200, 300, 400, 500, 590	10, 12, 14, 16, 18, 20, 20, 20	5, 8, 10, 12, 15, 18, 20, 20	–
C09–C16	GSFM	60, 100, 150, 200, 300, 400, 500, 590	20, 18, 16, 14, 12, 10, 18, 20	5, 8, 10, 12, 15, 18, 20, 20	–
C17–C19	Phase-coded	60, 100, 150	10, 12, 14	0.7000, 0.5833, 0.5000	Barker-7
C20–C24	Phase-coded	200, 300, 400, 500, 590	16, 18, 20, 18, 20	0.8125, 0.7222, 0.6500, 0.7222, 0.6500	Barker-13

Table 4: Compact representation of the exact 24-candidate parameter set.

For phase-coded candidates, the implementation enforces

$$T_{\text{chip}} = \frac{1}{B}, \quad T_p = \frac{N}{B}, \quad BT_p = N, \quad (5)$$

where N is the Barker length. This keeps chip duration, pulse duration and bandwidth consistent.

3.3 Baseband waveform generation

For LFM the implementation uses

$$x_{\text{LFM}}(t) = \exp\left[j\pi \frac{B}{T_p} t^2\right]. \quad (6)$$

For GSFM, the frequency range $[-B/2, B/2]$ is weighted by

$$w(f) = 0.54 + 0.46 \cos\left(\frac{2\pi f}{B}\right), \quad (7)$$

then mapped through a cumulative group-delay curve to obtain the nonlinear instantaneous-frequency trajectory. For phase coding, the Barker ± 1 sequence is repeated with the required samples-per-chip.

3.4 Waveform quality from the actual signal

The final generator samples each baseband waveform and derives autocorrelation metrics. The project therefore uses calculated sidelobe values rather than labels or hard-coded numbers.

4. Acoustic Propagation Model

4.1 Sound speed and pressure proxy

The supplied Mackenzie sound-speed field is used as the local sound speed c . A simple depth-to-pressure proxy in the generator is

$$P_{\text{dbar}} \approx \frac{D_{\text{m}}}{10.1}. \quad (8)$$

Turbidity is deliberately kept out of the Mackenzie sound-speed relation and is introduced later through reverberation.

4.2 François–Garrison absorption

The final fixed generator uses François–Garrison absorption rather than the earlier Thorp approximation. With f in kHz and $\theta = 273 + T$,

$$\alpha(f) = \frac{A_1 f_1 f^2}{f_1^2 + f^2} + \frac{A_2 P_2 f_2 f^2}{f_2^2 + f^2} + A_3 P_3 f^2. \quad (9)$$

The implemented terms are

$$A_1 = \frac{8.86}{c} 10^{0.78 \text{pH} - 5}, \quad f_1 = 2.8 \sqrt{\frac{S}{35}} 10^{4 - 1245/\theta}, \quad (10)$$

$$A_2 = \frac{21.44 S}{c} (1 + 0.025 T), \quad P_2 = 1 - 1.37 \times 10^{-4} D + 6.2 \times 10^{-9} D^2, \quad (11)$$

$$f_2 = \frac{8.17 \times 10^{8 - 1990/\theta}}{1 + 0.0018(S - 35)}, \quad P_3 = 1 - 3.83 \times 10^{-5} D + 4.9 \times 10^{-10} D^2, \quad (12)$$

with the temperature-dependent A_3 polynomial implemented exactly as in the final generator. The key design effect is that α is evaluated at each candidate frequency. Hence two otherwise similar waveforms can have different propagation loss because they use different carriers.

4.3 Transmission loss and received level

For range R in metres,

$$TL_k = 20 \log_{10}(R) + \alpha_k \frac{R}{1000}. \quad (13)$$

The candidate source level is

$$SL_k = 180 + 20 \log_{10}(A_k), \quad (14)$$

with normalized amplitude $A_k = 0.80$ for LFM/GSFM and 0.75 for phase-coded candidates. The active-sonar received level is

$$RL_k = SL_k - 2 TL_k + TS. \quad (15)$$

The two-way loss is important because the transmitted signal travels to the target and the echo returns to the receiver.

Design consequence: candidate frequency is part of the optimization, not only a waveform label, because absorption changes with frequency.

5. Noise, Turbidity, Reverberation and Detection

5.1 Frequency-dependent ambient noise

The final generator uses thermal and wind components. For $f_k = f/1000$ (f in Hz),

$$N_{\text{thermal}} = -15 + 20\log_{10}(f_k), \quad N_{\text{wind}} = W_{1k} - 17\log_{10}(f_k), \quad (16)$$

where W_{1k} is sampled from 45–65 dB. The combined PSD is formed in the linear domain:

$$N_{\text{PSD}} = 10\log_{10}(10^{N_{\text{thermal}}/10} + 10^{N_{\text{wind}}/10}). \quad (17)$$

Noise over the candidate bandwidth is then

$$NL_{B,k} = N_{\text{PSD}}(f_{c,k}) + NF_{\text{rx}} + 10\log_{10}(B_k), \quad (18)$$

with $NF_{\text{rx}} = 10$ dB.

5.2 Turbidity through volume reverberation

Turbidity does not enter the sound-speed equation. Instead, the final model uses a volume-reverberation proxy:

$$S_v = S_{v,\text{ref}} + 10\log_{10}(\tau_{\text{turb}}) + 10\gamma\log_{10}\left(\frac{f}{f_{\text{ref}}}\right), \quad (19)$$

with

$$S_{v,\text{ref}} = -85 \text{ dB}, \quad \gamma = 3, \quad f_{\text{ref}} = 100 \text{ kHz}.$$

The pulse-compression cell is

$$\Delta R_{\text{cell}} = \frac{c}{2B}, \quad (20)$$

and the volume term is

$$V_{\text{dB}} = 10\log_{10}\left(\frac{c}{2B}\right) - 15 + 20\log_{10}(R). \quad (21)$$

The reverberation level becomes

$$Rev_k = SL_k - 2TL_k + S_{v,k} + V_{\text{dB},k}. \quad (22)$$

These scattering quantities are model-derived proxies, not measured coefficients.

5.3 Processing gain and detection

Matched-filter processing gain is approximated from the time-bandwidth product:

$$G_{p,k} \approx 10\log_{10}(B_k T_{p,k}). \quad (23)$$

Noise-limited processed performance:

$$SNR_{n,k} = RL_k - NL_{B,k} + G_{p,k}. \quad (24)$$

Signal-to-reverberation ratio:

$$SRR_k = RL_k - Rev_k. \quad (25)$$

The combined detection metric is

$$SINR_k = -10\log_{10}(10^{-SNR_{n,k}/10} + 10^{-SRR_k/10}). \quad (26)$$

Detection margin is

$$M_k = SINR_k - 10 \text{ dB}. \quad (27)$$

A candidate is detection-feasible when $M_k \geq 0$.

6. Range Resolution, Correlation, Sidelobes and Hardware Limits

6.1 Range resolution

Theoretical range resolution is

$$\Delta R_k = \frac{c}{2B_k}. \quad (28)$$

For each scenario the allowed maximum is sampled from 0.04–0.08 m, and the candidate must satisfy $\Delta R_k \leq \Delta R_{\max}$. This shows the main bandwidth trade-off: more bandwidth improves resolution but changes noise bandwidth, processing gain and hardware feasibility.

6.2 Correlation estimate

The final generator first computes pre-compression in-band linear SNR:

$$SNR_{\text{in,lin}} = 10^{(RL - NL_B)/10}, \quad (29)$$

then uses

$$\rho = \sqrt{\frac{SNR_{\text{in,lin}}}{1 + SNR_{\text{in,lin}}}}. \quad (30)$$

This is an analytical/model-derived correlation estimate, not a measured receive waveform correlation.

6.3 PSL and ISL

For sampled baseband $x[n]$, the implementation forms the full autocorrelation. If A_0 is the main peak and A_s is the largest sidelobe,

$$PSL = 20 \log_{10} \left(\frac{A_s}{A_0} \right). \quad (31)$$

Using the implementation-defined main-lobe boundary,

$$ISL = 10 \log_{10} \left(\frac{E_{\text{side}}}{E_{\text{main}}} \right). \quad (32)$$

More negative values mean lower sidelobes under this convention.

6.4 Energy proxy

The generator uses

$$E_{\text{proxy}} = A^2 T_p, \quad (33)$$

and requires $E_{\text{proxy}} \leq 1$ J. It is intentionally an engineering proxy, not calibrated electrical energy. A future hardware model can replace it with

$$E_J = \int V(t) I(t) dt \quad (34)$$

plus measured amplifier/transducer efficiency.

6.5 Hardware constraints

$$50 \leq f_c \leq 600 \text{ kHz}, \quad 10 \leq B \leq 20 \text{ kHz}, \quad (35)$$

$$E \leq 1 \text{ J}, \quad f_{\text{DAC}} \geq 2 \left(f_c + \frac{B}{2} \right), \quad (36)$$

with $f_{\text{DAC}} = 2$ MHz. The sampling-rate check is especially important for high-frequency candidates; it prevents a candidate that looks good acoustically from being selected when the digital generation chain cannot represent it correctly.

7. Constrained Multi-Objective Optimization

7.1 Feasibility first

OJAS does not score every candidate equally. It first checks

$$VALID_k = HWOK_k \wedge DETECTOK_k \wedge RESOK_k. \quad (37)$$

Only valid candidates enter the ranking stage.

7.2 Normalized score terms

The final generator uses the following normalized terms:

$$S_{\text{res},k} = \text{clip}\left(\frac{\min_j \Delta R_j}{\Delta R_k}, 0, 1\right), \quad (38)$$

$$S_{\text{PSL},k} = \text{clip}\left(\frac{-PSL_k}{40}, 0, 1\right), \quad S_{\text{ISL},k} = \text{clip}\left(\frac{-ISL_k}{40}, 0, 1\right), \quad (39)$$

$$S_{\text{margin},k} = \frac{\text{clip}(M_k, 0, 10)}{10}, \quad P_{E,k} = \text{clip}\left(\frac{E_k}{1}, 0, 1\right). \quad (40)$$

The sidelobe terms are activated according to reverberation severity:

$$N_{\text{side},k} = \text{clip}\left(\frac{25 - SRR_k}{25}, 0, 1\right). \quad (41)$$

Thus, sidelobe quality matters more when signal and reverberation are closer.

7.3 Final engineering score

The implemented score is

$$J_k = 0.30 S_{\text{res},k} + N_{\text{side},k}(0.30 S_{\text{PSL},k} + 0.15 S_{\text{ISL},k}) + 0.25 S_{\text{margin},k} - 0.15 P_{E,k}. \quad (42)$$

The selected candidate is

$$k^* = \arg \max_{k: VALID_k} J_k. \quad (43)$$

7.4 Fallback when nothing is feasible

The final implementation fixes a common edge case: if every candidate fails the full feasibility test, a blind argmax over invalid scores must not silently return C01. Instead,

$$k_{\text{fb}} = \arg \max_{k: HWOK_k} M_k. \quad (44)$$

The output records either `optimal_valid` or `fallback_max_margin`.

7.5 Why this structure matters

The score balances four practical goals: good resolution, low sidelobes when reverberation matters, adequate detection margin and low energy. Because hard limits are checked first, a strong score in one area cannot override an invalid operating point.

8. Offline Dataset, LUT and Runtime Decision

8.1 What the final dataset stores

For each selected scenario, the final generated dataset stores the source environment together with model outputs including:

- selected candidate ID, waveform family, frequency, bandwidth, pulse duration and amplitude;
- absorption, one-way and two-way transmission loss, received level and noise level;
- reverberation, processing gain, combined SINR metric and detection margin;
- theoretical range resolution, PSL, ISL, correlation and estimated energy;
- hardware/detection/resolution flags, selection status, feasible-count and top-ranked alternatives.

The project manifest identifies the final generated dataset as a 22,068-row scenario file and separately identifies the offline candidate table, the W01–W24 selection LUT and the final 24-entry waveform library.

8.2 Why the LUT is a compression step

The full offline problem is

$$(1,839 \times 12)_{\text{scenarios}} \times 24_{\text{candidates}} = 529,632$$

evaluations. The embedded runtime should not repeat all of these calculations. Instead, the selected result is compressed into a small deployment representation: environment → LUT → W01–W24 → waveform (Figure 1).

8.3 Deployment-side mapping

The supplied artifacts define three key deployment files:

```
OJAS_FINAL_SENSOR_MACKENZIE_LUT_24.csv
OJAS_final_waveform_library_24_Barker.csv
OJAS_waveform_debug_Barker_FINAL.m
```

The exact threshold table for the W01–W24 mapping is not exposed in the uploaded document text, so this report intentionally does not invent threshold values. It preserves the documented interface: environmental state to Wxx identifier, then Wxx to a waveform definition.

8.4 Offline-to-runtime boundary

The expensive part is offline: propagation, noise, reverberation, candidate generation and scoring. The runtime part is compact: receive environmental values, obtain the mapped waveform ID and synthesize/use the stored waveform definition.

9. MATLAB Validation and Demonstration

9.1 Deployment-side validation

The final MATLAB debugger reads the ocean dataset, sensor/Mackenzie LUT and waveform library rather than hard-coding the mapping inside the viewer. This keeps the demonstration tied to the supplied deployment artifacts. For LFM and nonlinear-FM waveforms, the generated instantaneous frequency is compared with the theoretical trajectory. For phase-coded signals, the carrier and phase-code sequence are checked. The project documentation also notes that finite-record Hilbert-edge effects are excluded from the numerical instantaneous-frequency check for phase-coded records.

9.2 Ten-second random-ocean replay

The reproducible live demonstration repeatedly chooses a valid ocean row every ten seconds:

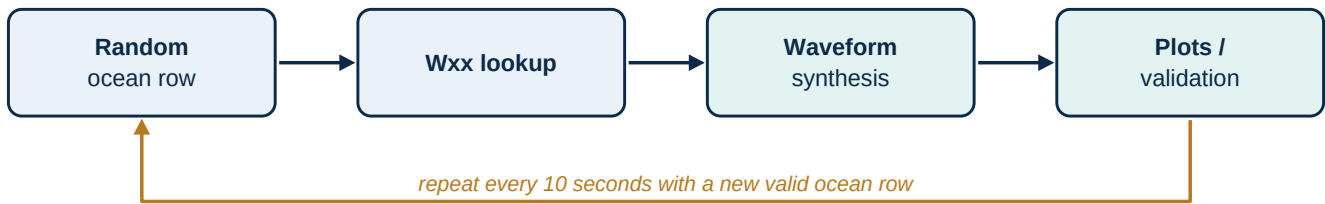


Figure 3: MATLAB ten-second random-ocean replay loop.

The display shows the environmental state, Mackenzie sound speed, selected Wxx, waveform family, waveform and spectrum. The purpose is to make the adaptive decision visible as the environment changes.

9.3 What is demonstrated versus what is measured

The project status file states that the reproducible live demonstration is the MATLAB 10-second replay. The MATLAB waveform plots are numerical synthesis/validation outputs. They are not described as measured DAC or oscilloscope captures unless separate hardware evidence is available.

9.4 Reproducibility chain

The relevant artifacts listed by the project are:

- 1 final fixed Python dataset generator;
- 2 final ocean input and generated scenario dataset;
- 3 offline candidate table and W01–W24 LUT;
- 4 final waveform library;
- 5 MATLAB random-row demo and waveform debugger;
- 6 Arduino environmental emitter;
- 7 STM32 firmware and CubeMX handover packages;
- 8 methodology/provenance and demonstration-status files.

The same source data and LUT definitions are used across the numerical validation path.

10. Embedded Architecture, Status and Limits

10.1 Intended embedded path

The documented runtime architecture is Arduino environmental input → STM32 processing → laptop monitoring. The Arduino acts as the environmental-data front end for the prototype. The STM32 is the intended real-time processing and waveform-generation platform. The laptop is used as the monitoring and validation interface.

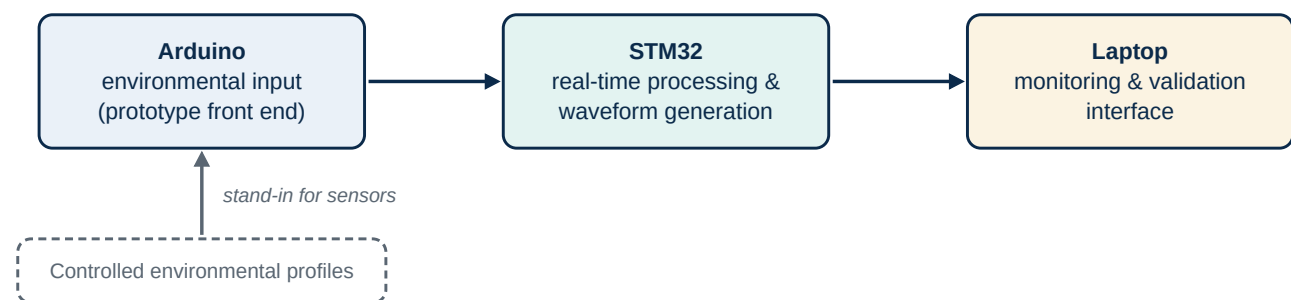


Figure 4: Intended embedded runtime path of the OJAS prototype.

10.2 Current prototype condition

The supplied documentation states that the prototype does not yet contain the complete oceanographic sensor suite. Controlled environmental profiles are therefore used to reproduce sensor input. This lets the data contract and decision path be tested before replacing the profiles with calibrated physical sensors.

10.3 What is already completed

The supplied demonstration-status file marks the following as completed and reproducible:

- real ocean dataset ingestion;
- offline 24-candidate acoustic evaluation;
- model-based dataset generation;
- W01–W24 deployment-side LUT selection;
- MATLAB waveform synthesis and validation;
- 10-second random-ocean replay;
- Arduino environmental packet emitter;
- STM32 firmware source and CubeMX materials.

10.4 Model boundaries

The current implementation is an analytical engineering model. The following should therefore be read as estimates unless independently measured: absorption, propagation loss, received level, ambient noise, reverberation, detection metric, correlation and energy.

The reverberation model uses tunable proxy constants. Source level and receiver noise figure are assumptions. The energy model is a normalized amplitude-duration proxy. The final validated dataset generator uses the analytical model documented in the repository; Bellhop was considered as a higher-fidelity future propagation layer, but no validated Bellhop result is claimed without an archived run.

10.5 Practical next steps

A future calibrated system can replace the proxies with measured transducer response, amplifier behavior, receiver noise, physical scattering data and more detailed propagation modeling without changing the basic OJAS decision structure.

11. Implementation Workflow and Conclusion

11.1 Complete algorithm

```

FOR each ocean observation
  FOR each of 12 target ranges
    sample target strength, wind noise, resolution requirement
    FOR each candidate C01...C24
      synthesize baseband waveform
      compute PSL, ISL and energy proxy
      compute absorption and transmission loss
      compute received level, noise and reverberation
      compute processing gain, SINR and detection margin
      compute range resolution and correlation
      check hardware, detection and resolution constraints
    END
    select best feasible candidate, or hardware-legal fallback
    store metrics and decision
  END
END

```

Algorithm 1: Complete offline evaluation and selection procedure.

11.2 Main contribution

The main contribution of OJAS is the connection between environmental awareness and waveform design. The system does not choose a waveform because it is universally “best”. It chooses a waveform after testing how the waveform behaves under the current environment and scenario. The central chain is

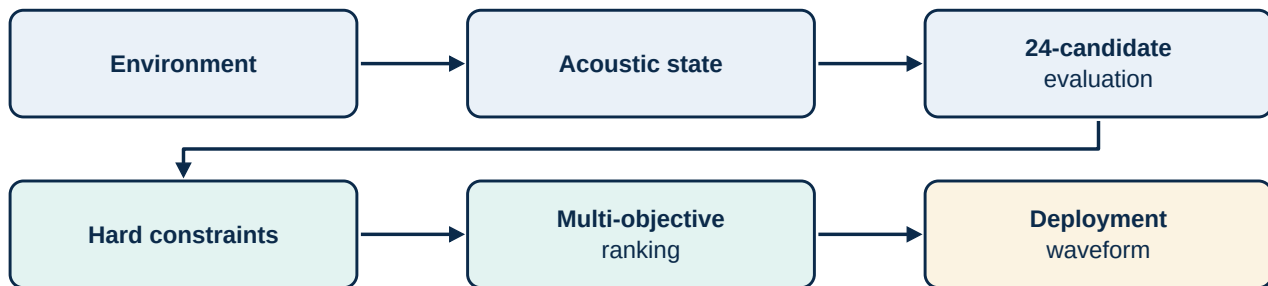


Figure 5: The central OJAS chain.

The solution is built around four useful principles:

- 1 **Environment-aware choice:** frequency, bandwidth and pulse shape are allowed to change with the operating condition.
- 2 **Physics-based filtering:** absorption, propagation, noise and reverberation are included before the decision.
- 3 **Feasibility before scoring:** an acoustically attractive but hardware-invalid candidate cannot win.
- 4 **Offline computation, compact runtime:** the large search is completed offline and compressed into a deployable lookup structure.

11.3 Final conclusion

OJAS starts with approximately 1,839 supplied ocean observations, creates 22,068 environment–range scenarios and evaluates 24 waveform candidates per scenario, giving 529,632 candidate evaluations. The final fixed generator uses François–Garrison absorption, frequency-dependent noise, turbidity-dependent volume reverberation, matched-filter processing gain, combined SINR-style detection performance, theoretical range resolution, sampled-waveform PSL/ISL and explicit hardware constraints.

The result is not a single static waveform. It is a repeatable method for deciding which waveform should be transmitted under a given condition, followed by a compact W01–W24 deployment representation and MATLAB validation path.

The environment defines the acoustic state; the acoustic state determines which waveform is worth transmitting.

11.4 Source artifacts used

generate_ocean_acoustic_dataset.py; METHODOLOGY_AND_PROVENANCE.md;
 OJAS_Final_Judges_Presentation_YouTube_Script.docx; FILE_MANIFEST.md; DEMO_STATUS.md;
 SIH_2026_dataset_1.ipynb.

The final fixed generator is treated as authoritative for the final implementation where earlier notebook versions differ.